

|                                                     |                                              |                                               |                                           |                                                     |                                           |                                               |
|-----------------------------------------------------|----------------------------------------------|-----------------------------------------------|-------------------------------------------|-----------------------------------------------------|-------------------------------------------|-----------------------------------------------|
|                                                     | $\mathcal{K}_{0^+}^{\#1}$                    | $\mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$ |                                           |                                                     |                                           |                                               |
| $\mathcal{K}_{0^+}^{\#1} \dagger$                   | 0                                            | 0                                             |                                           |                                                     |                                           |                                               |
| $\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$ | 0                                            | $3k^2 \frac{(4)}{\kappa_{15}}$                | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$           | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$      | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$          |
| $\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$     | $-\frac{k^2 \frac{(4)}{\kappa_1}}{6}$        | $\frac{k^2 \frac{(4)}{\kappa_1}}{3\sqrt{2}}$  | 0                                         | 0                                                   |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$     | $\frac{k^2 \frac{(4)}{\kappa_1}}{3\sqrt{2}}$ | $-\frac{k^2 \frac{(4)}{\kappa_1}}{3}$         | 0                                         | 0                                                   |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha}$          | 0                                            | 0                                             | 0                                         | 0                                                   |                                           |                                               |
| $\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha}$          | 0                                            | 0                                             | 0                                         | 0                                                   | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |
|                                                     |                                              |                                               |                                           | $\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$     | $\frac{3k^2 \frac{(4)}{\kappa_1}}{2}$     | 0                                             |
|                                                     |                                              |                                               |                                           | $\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$ | 0                                         | 0                                             |

|                                                     |                                               |                                               |                                           |                                                     |                                           |                                               |
|-----------------------------------------------------|-----------------------------------------------|-----------------------------------------------|-------------------------------------------|-----------------------------------------------------|-------------------------------------------|-----------------------------------------------|
|                                                     | $\mathcal{J}_{0^+}^{\#1}$                     | $\mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi}$ |                                           |                                                     |                                           |                                               |
| $\mathcal{J}_{0^+}^{\#1} \dagger$                   | 0                                             | 0                                             |                                           |                                                     |                                           |                                               |
| $\mathcal{J}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$ | 0                                             | $\frac{1}{3k^2 \frac{(4)}{\kappa_{15}}}$      | $\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta}$           | $\mathcal{J}_{1^+}^{\#1}{}_{\alpha}$      | $\mathcal{J}_{1^+}^{\#2}{}_{\alpha}$          |
| $\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta}$     | $-\frac{2}{3k^2 \frac{(4)}{\kappa_1}}$        | $\frac{2\sqrt{2}}{3k^2 \frac{(4)}{\kappa_1}}$ | 0                                         | 0                                                   |                                           |                                               |
| $\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta}$     | $\frac{2\sqrt{2}}{3k^2 \frac{(4)}{\kappa_1}}$ | $-\frac{4}{3k^2 \frac{(4)}{\kappa_1}}$        | 0                                         | 0                                                   |                                           |                                               |
| $\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha}$          | 0                                             | 0                                             | 0                                         | 0                                                   |                                           |                                               |
| $\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha}$          | 0                                             | 0                                             | 0                                         | 0                                                   | $\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |
|                                                     |                                               |                                               |                                           | $\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta}$     | $\frac{2}{3k^2 \frac{(4)}{\kappa_1}}$     | 0                                             |
|                                                     |                                               |                                               |                                           | $\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$ | 0                                         | 0                                             |

### Lagrangian

$$\begin{aligned}
& \frac{(4)}{\kappa_1} \partial_\beta \mathcal{K}_{\chi\delta}^\delta \partial^\chi \mathcal{K}_\alpha^{\alpha\beta} - \frac{4}{3} \frac{(4)}{\kappa_1} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta + 2 \frac{(4)}{\kappa_{15}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta + \\
& \frac{2}{3} \frac{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - 4 \frac{(4)}{\kappa_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \frac{5}{3} \frac{(4)}{\kappa_1} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - 2 \frac{(4)}{\kappa_{15}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - \\
& \frac{1}{3} \frac{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \frac{(4)}{\kappa_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \frac{(4)}{\kappa_{15}} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \frac{(4)}{\kappa_{15}} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

|                                                                                  |                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added source term(s):                                                            | $\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Source constraint(s)                                                             | # constraint(s)                                               | Covariant form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $\mathcal{J}_{0^+}^{\#1} == 0$                                                   | 1                                                             | $\partial_\beta \mathcal{J}_\alpha^{\alpha\beta} == 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| $\mathcal{J}_{1^+}^{\#2\alpha} == 0$                                             | 3                                                             | $\partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $\mathcal{J}_{1^+}^{\#1\alpha} == 0$                                             | 3                                                             | $\partial_\chi \partial^\alpha \mathcal{J}_\beta^{\beta\chi} + \partial_\chi \partial^\chi \mathcal{J}_\beta^{\beta\alpha} == \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| $2 \mathcal{J}_{1^+}^{\#1\alpha\beta} + \mathcal{J}_{1^+}^{\#2\alpha\beta} == 0$ | 3                                                             | $\partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\chi \mathcal{J}^{\chi\alpha\beta} == \partial_\chi \mathcal{J}^{\beta\alpha\chi}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta\chi} == 0$                                    | 5                                                             | $ \begin{aligned} & 3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \\ & 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}_\delta{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\alpha}_\delta == \\ & 3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \\ & 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}_\delta{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\beta}_\delta \end{aligned} $ |
| Total # constraint(s):                                                           | 15                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Resolved unitarity condition(s):                                                 | True                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |